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22-05-2026

TO WHOM IT MAY CONCERN

This is to certify that **Mr. Muhammad Haris**, S/O Waheed Iqbal, has been serving as a **Research Assistant** at the Functional Genomics & Tissue Culture Laboratory, Department of Bioinformatics and Biotechnology, Government College University Faisalabad since **September 1, 2024**. During his tenure in our laboratory, he has worked with great dedication, professionalism, and technical competence in the fields of biotechnology, functional genomics, molecular biology, bioinformatics, and scientific software development.

Mr. Haris has gained hands-on experience in cell tissue culture techniques, DNA/RNA extraction, PCR, qPCR, gel electrophoresis, and other molecular biology procedures. Alongside his laboratory work, he has also worked extensively on computational biology and bioinformatics projects involving bulk RNA-Seq analysis, scRNA-Seq analysis, Whole Exome Sequencing (WES) analysis, phylogenetic analysis, functional annotation, pathway enrichment analysis, and Linux- and R-based bioinformatics workflows.

During his service in our laboratory, he successfully developed several scientific databases and bioinformatics platforms, including **HCMdrugs**, Pakistan's first dedicated web-based database for therapeutic options related to hypertrophic cardiomyopathy; **PGF-Explorer**, a database for gene family data exploration and analysis and **AutoRNAseq**, an automated RNA-Seq analysis pipeline. Furthermore, he has worked on five first-author research manuscripts spanning the fields of genomics and transcriptomics, reflecting his strong research aptitude and scientific writing abilities. Mr. Haris has actively represented our laboratory at several national and international conferences, symposiums, and scientific events, where he presented his research work and effectively communicated scientific findings to the academic community.

Throughout his tenure, Mr. Haris proved himself to be sincere, hardworking, innovative, and capable of working independently as well as within a research team.

We wish him success in all his future academic and professional endeavors.



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